

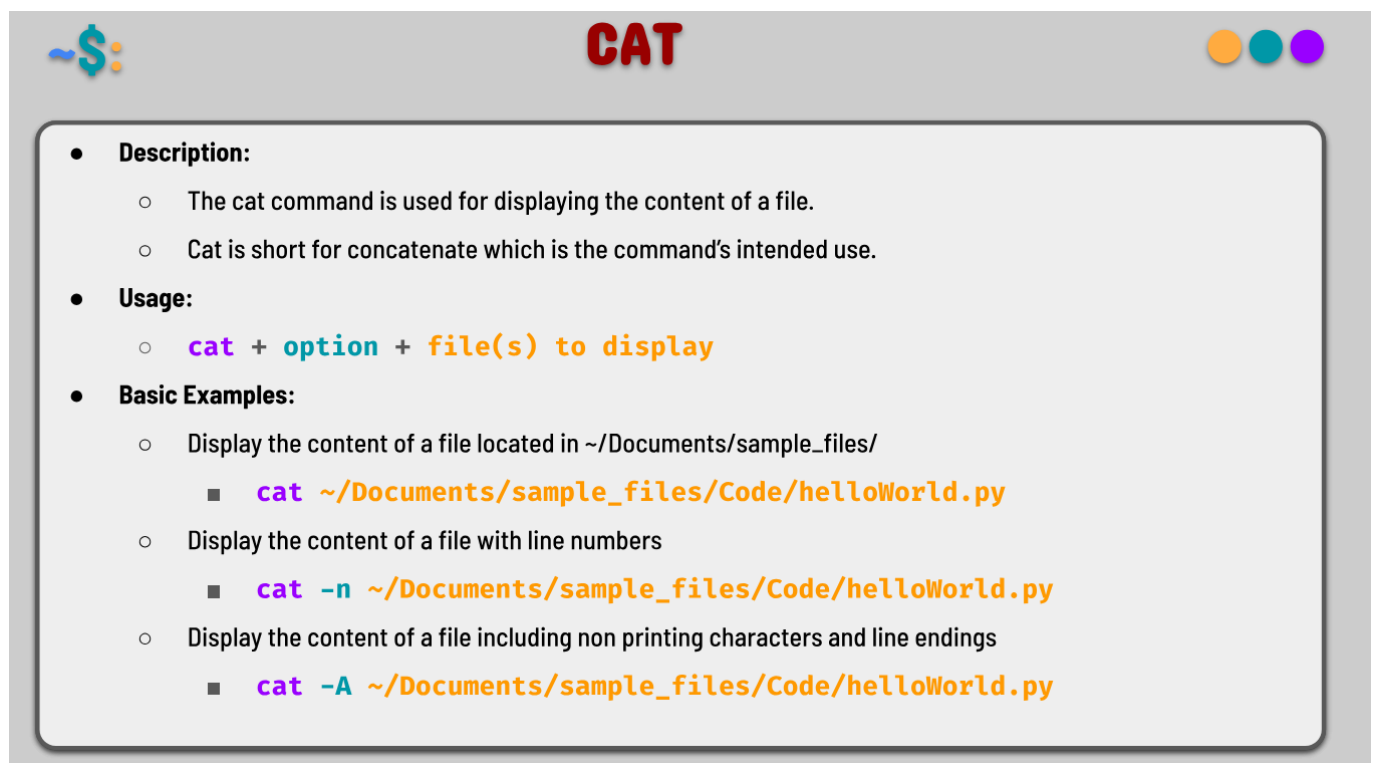
Notes 8

File and Text Processing Commands in Linux

cat

Command name: `cat`

- **Description:** Concatenates and displays the content of one or more files on the standard output (screen).
- **Formula/Syntax:** `cat [OPTION] [FILE]...`
- **Examples:** `cat file.txt` *Display content of a file*
`cat file1.txt file2.txt` *Display multiple files*
`cat -n file.txt` *Display content with line numbers*



The image shows a terminal window with a title bar containing a terminal icon, the word 'CAT' in large red letters, and three window control buttons (orange, teal, purple). The terminal content is as follows:

```
• Description:

- The cat command is used for displaying the content of a file.
- Cat is short for concatenate which is the command's intended use.

```

```
• Usage:

- cat + option + file(s) to display

```

```
• Basic Examples:

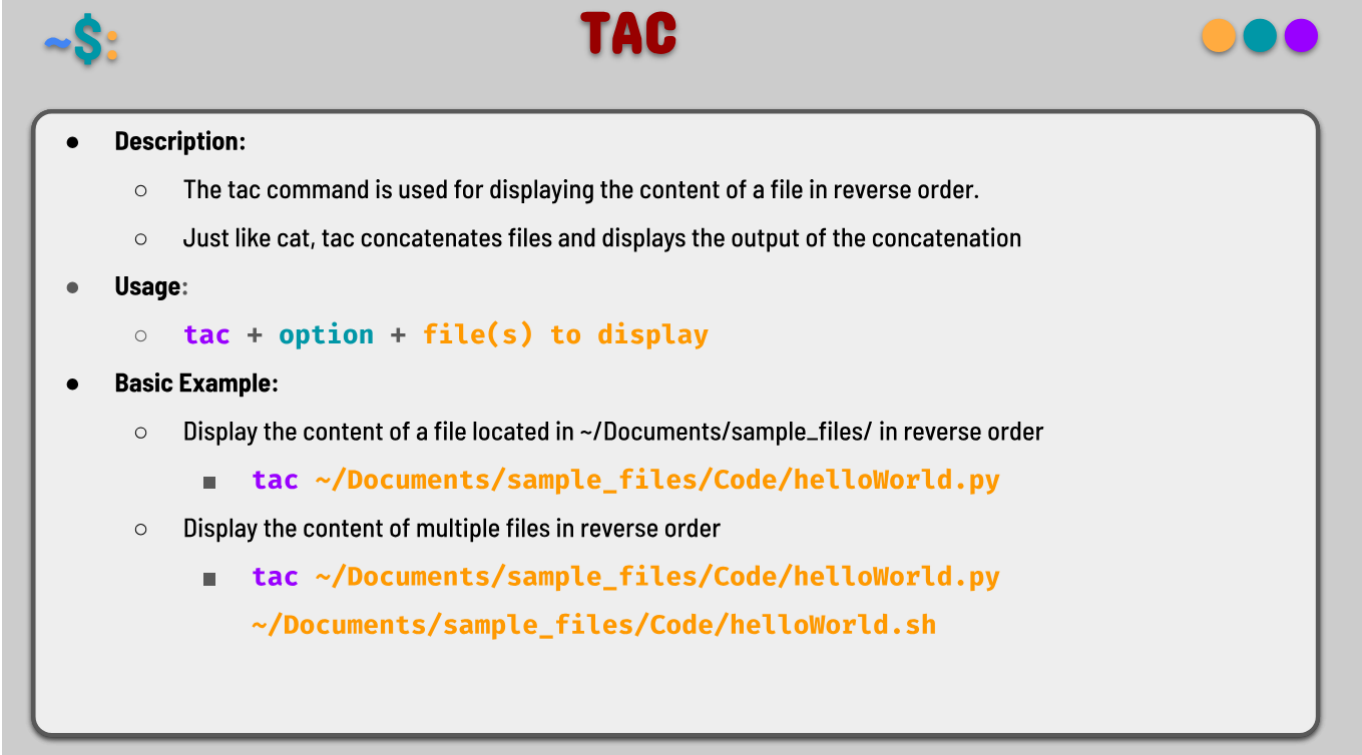
- Display the content of a file located in ~/Documents/sample_files/
  - cat ~/Documents/sample_files/Code/helloWorld.py
- Display the content of a file with line numbers
  - cat -n ~/Documents/sample_files/Code/helloWorld.py
- Display the content of a file including non printing characters and line endings
  - cat -A ~/Documents/sample_files/Code/helloWorld.py

```

tac

Command name: `tac`

- **Description:** Concatenates and prints files in reverse order (from last line to first).
- **Formula/Syntax:** `tac [OPTION] [FILE]...`
- **Examples:** `tac file.txt` *Display file content in reverse order*

A terminal window titled "TAC" with a terminal icon on the left and three colored window control buttons (orange, teal, purple) on the right. The window contains a list of information about the tac command.

- **Description:**
 - The tac command is used for displaying the content of a file in reverse order.
 - Just like cat, tac concatenates files and displays the output of the concatenation
- **Usage:**
 - `tac + option + file(s) to display`
- **Basic Example:**
 - Display the content of a file located in ~/Documents/sample_files/ in reverse order
 - `tac ~/Documents/sample_files/Code/helloWorld.py`
 - Display the content of multiple files in reverse order
 - `tac ~/Documents/sample_files/Code/helloWorld.py`
`~/Documents/sample_files/Code/helloWorld.sh`

head

Command name: head

- **Description:** Outputs the first part (beginning) of a file. By default, it displays the first 10 lines.
- **Formula/Syntax:** head [OPTION] [FILE]...
- **Examples:** head file.txt *Show first 10 lines*
head -n 20 file.txt *Show first 20 lines*
head -c 100 file.txt *Show first 100 characters*



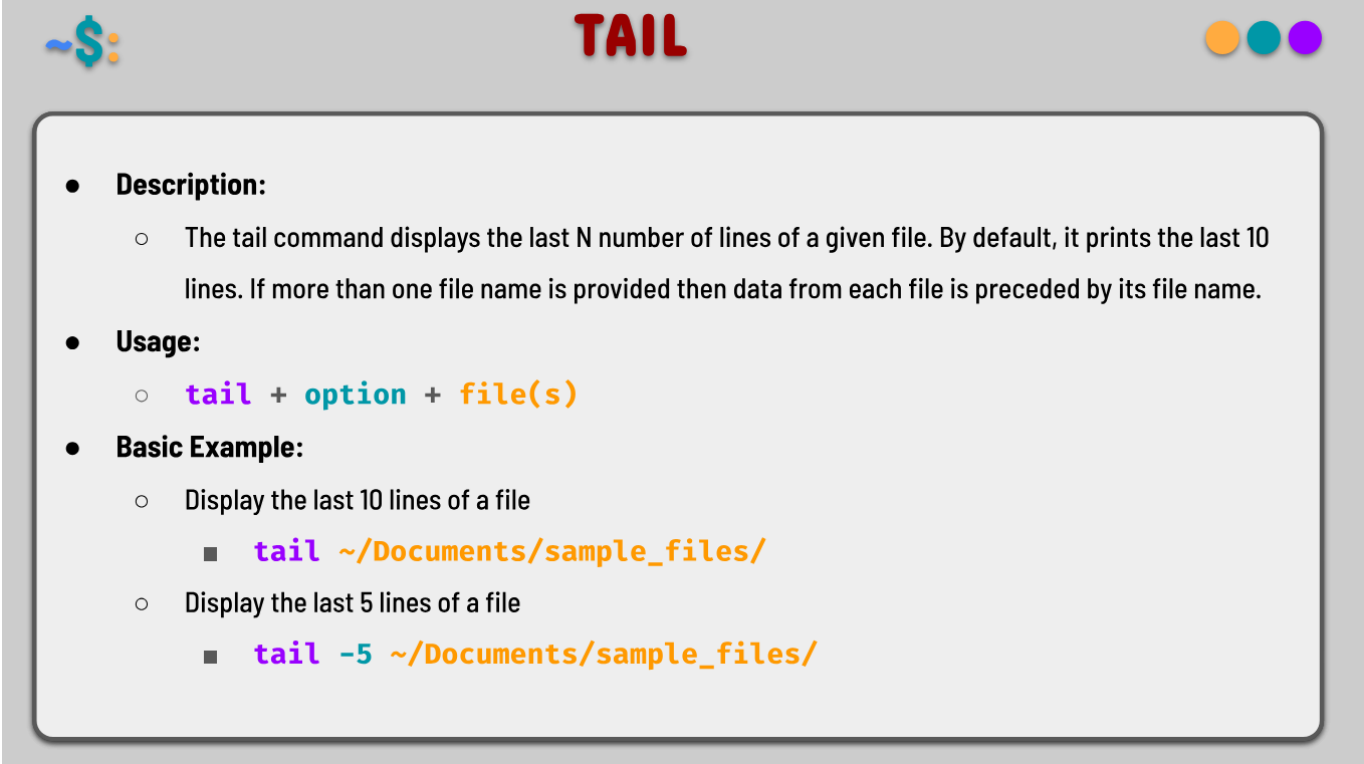
HEAD

- **Description:**
 - The head command displays the top N number of lines of a given file. By default, it prints the first 10 lines. If more than one file name is provided then data from each file is preceded by its file name.
- **Usage:**
 - `head + option + file(s)`
- **Basic Example:**
 - Display the first 10 lines of a file
 - `head ~/Documents/Book/dracula.txt`
 - Display the first 5 lines of a file
 - `head -5 ~/Documents/Book/dracula.txt`

tail

Command name: `tail`

- **Description:** Outputs the last part (end) of a file. By default, it displays the last 10 lines.
- **Formula/Syntax:** `tail [OPTION] [FILE]...`
- **Examples:** `tail file.txt` *Show last 10 lines*
`tail -n 50 file.txt` *Show last 50 lines*
`tail -f /var/log/apache2/error.log` *Follow log file in real-time*



- **Description:**
 - The tail command displays the last N number of lines of a given file. By default, it prints the last 10 lines. If more than one file name is provided then data from each file is preceded by its file name.
- **Usage:**
 - `tail + option + file(s)`
- **Basic Example:**
 - Display the last 10 lines of a file
 - `tail ~/Documents/sample_files/`
 - Display the last 5 lines of a file
 - `tail -5 ~/Documents/sample_files/`

cut

Command name: `cut`

- **Description:** Removes sections from each line of a file and outputs only selected portions (commonly used to extract columns or fields).
- **Formula/Syntax:** `cut [OPTION] [FILE]...`
- **Examples:** `cut -d':' -f1 /etc/passwd` *Extract usernames from /etc/passwd*
`cut -f1,3 file.txt` *Extract fields 1 and 3*
`cut -c1-10 file.txt` *Extract first 10 characters of each line*



The cut command



- **Description:**
 - The cut command is used to extract a specific section of each line of a file and display it to the screen.
- **Usage:**
 - `cut + option + file(s)`
- **Basic Example:**
 - Display a list of all the users in your system
 - `cut -d ':' -f1 /etc/passwd`
 - Display a list of all the users in your system with their login shell
 - `cut -d ':' -f1,7 /etc/passwd`

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sort

Command name: `sort`

- **Description:** Sorts lines of text files in ascending or descending order.
- **Formula/Syntax:** `sort [OPTION] [FILE]...`
- **Examples:** `sort file.txt` Sort lines alphabetically
`sort -n numbers.txt` Sort numerically
`sort -r file.txt` Sort in reverse order



SORT



- **Description:**
 - The sort command is used for sorting files. The sort command supports sorting: alphabetically, in reverse order, by number, and by month.
 - The sort command follows this order unless specified otherwise:
 - Lines starting with a number will appear before lines starting with a letter.
 - Lines starting with a letter that appears earlier in the alphabet will appear before lines starting with a letter that appears later in the alphabet.
 - Lines starting with a lowercase letter will appear before lines starting with the same letter in uppercase.
- **Usage:**
 - `sort + option + file`
- **Basic Example:**
 - Sort a file
 - `sort users.lst`

WC

Command name: `WC`

- **Description:** Prints the number of lines, words, and characters in a file or from standard input.
- **Formula/Syntax:** `WC [OPTION] [FILE]...`
- **Examples:** `WC file.txt` *Show lines, words, and characters*
`WC -l file.txt` *Count lines only*
`WC -w file.txt` *Count words only*



WC



- **Description:**
 - The wc command is used for printing the number of lines, characters and bytes in a file
- **Usage:**
 - `wc + option + file(s)`
- **Basic Example:**
 - Display the number of characters in a file
 - `wc -m users.txt`
 - Display the number of lines in a file
 - `wc -l users.txt`
 - Display the number words in a file
 - `wc -w users.txt`